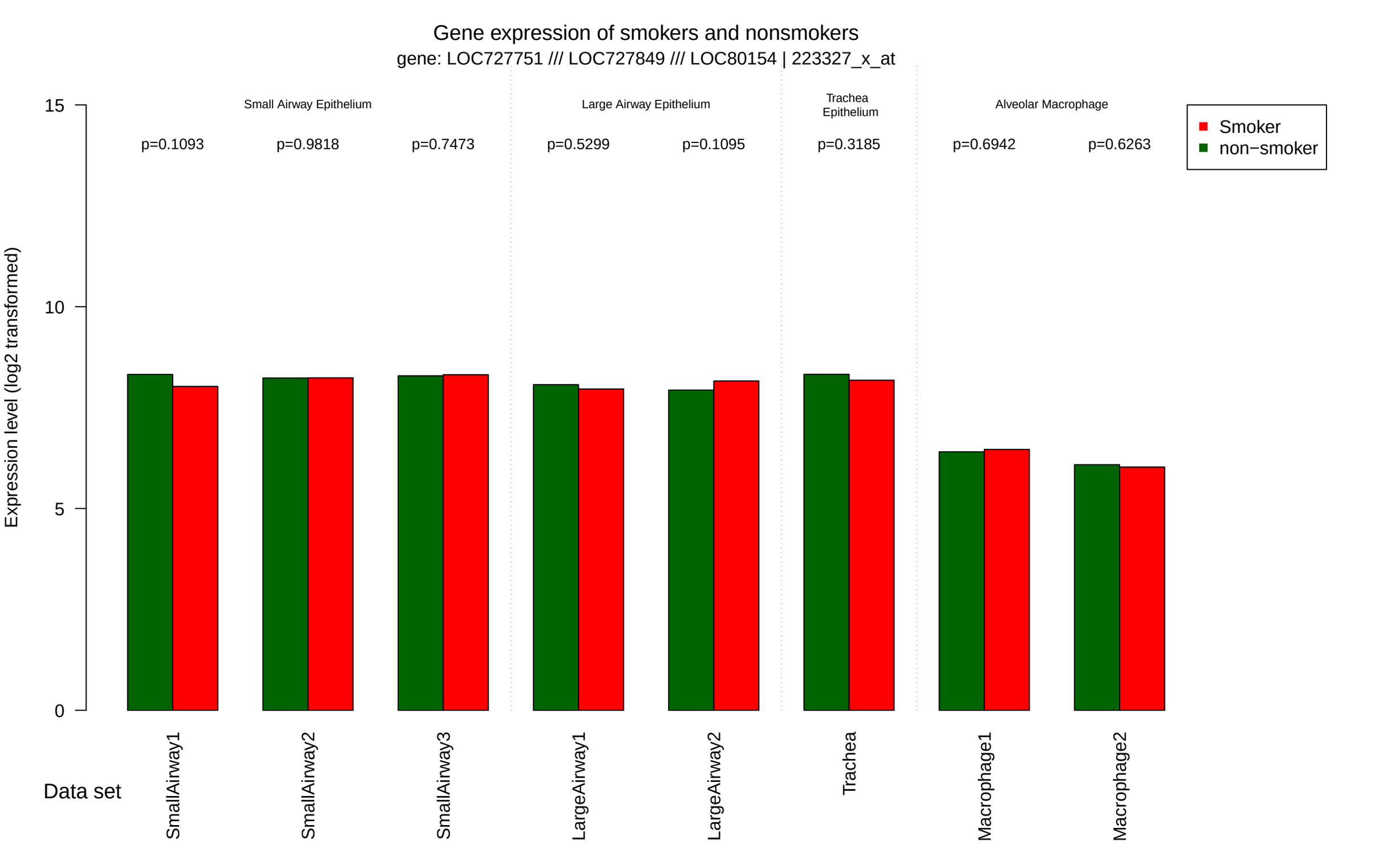
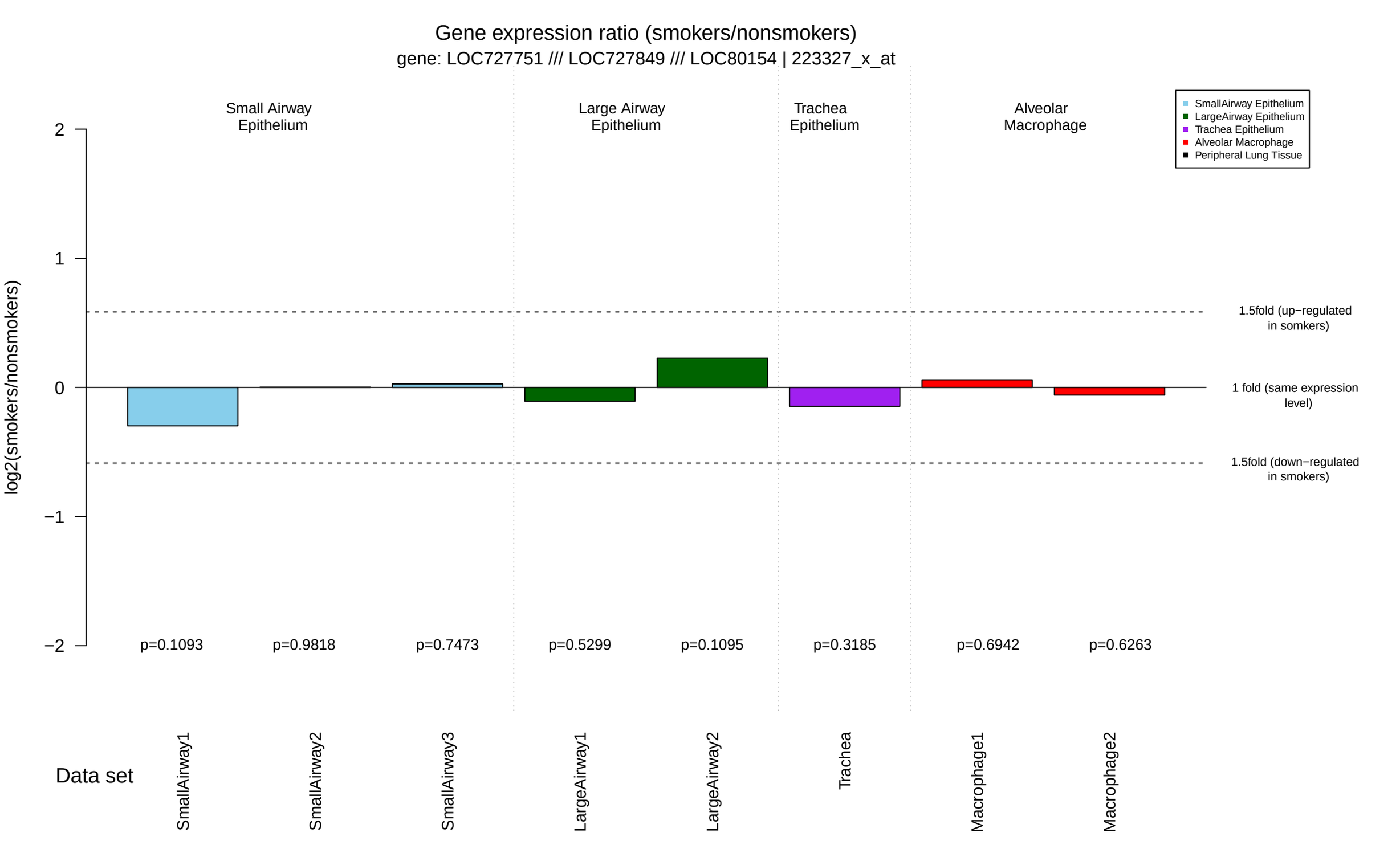
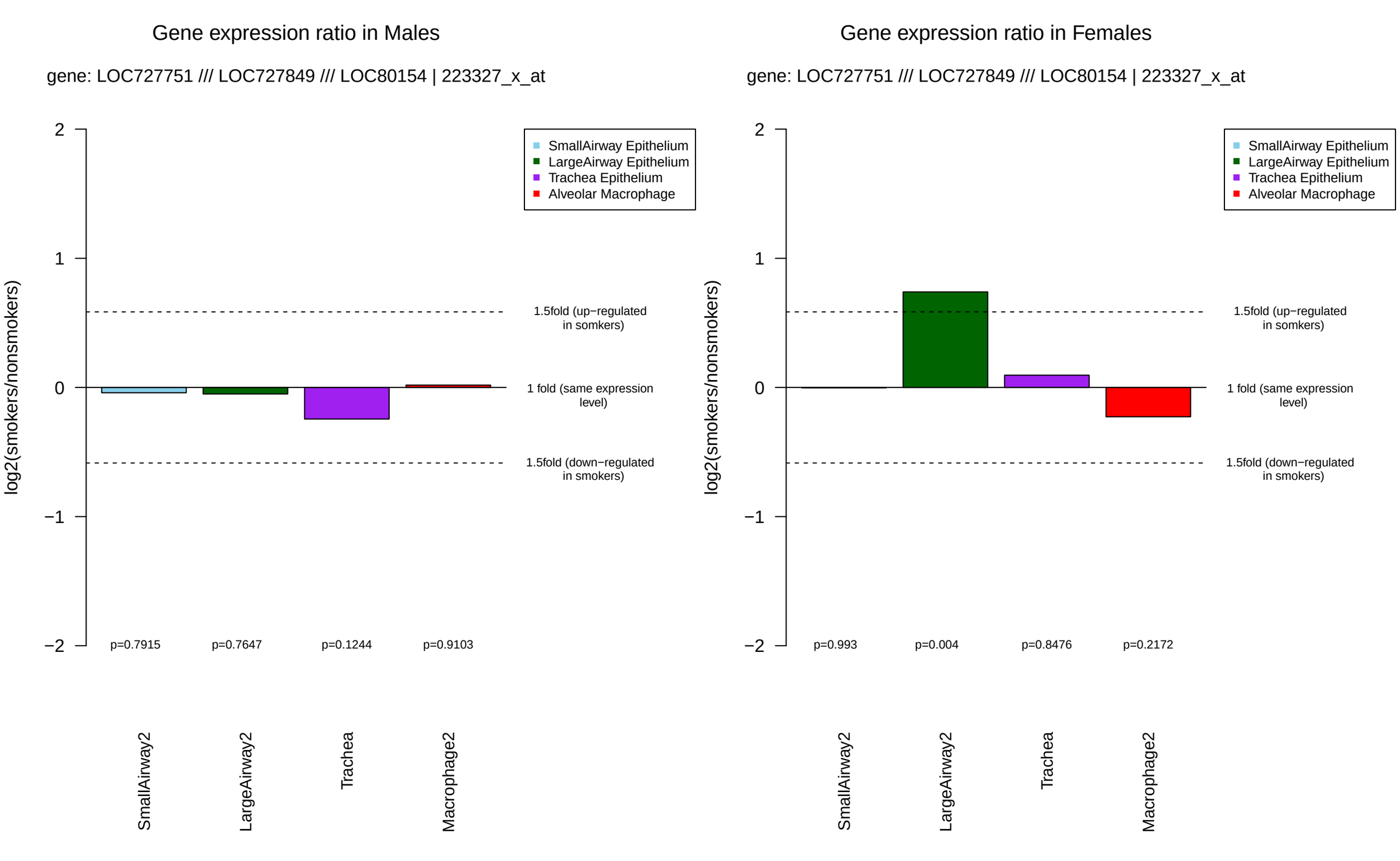
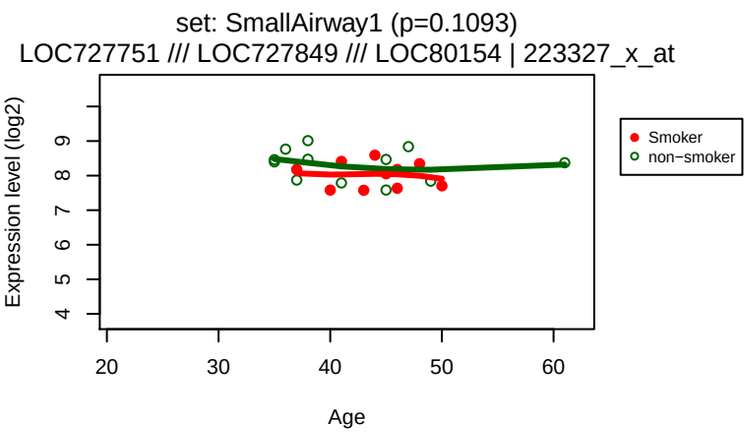
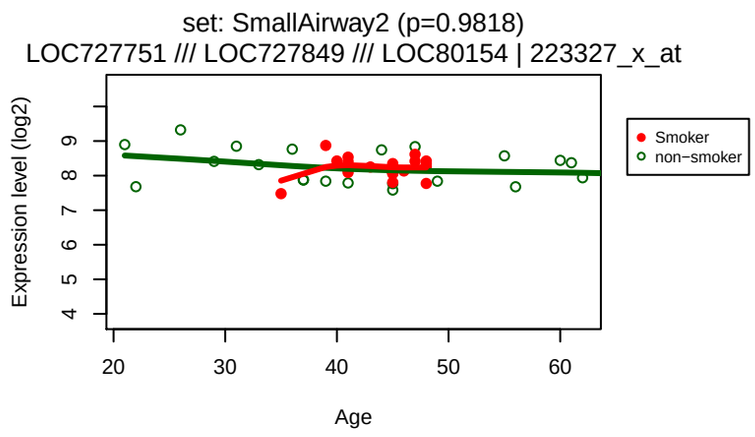
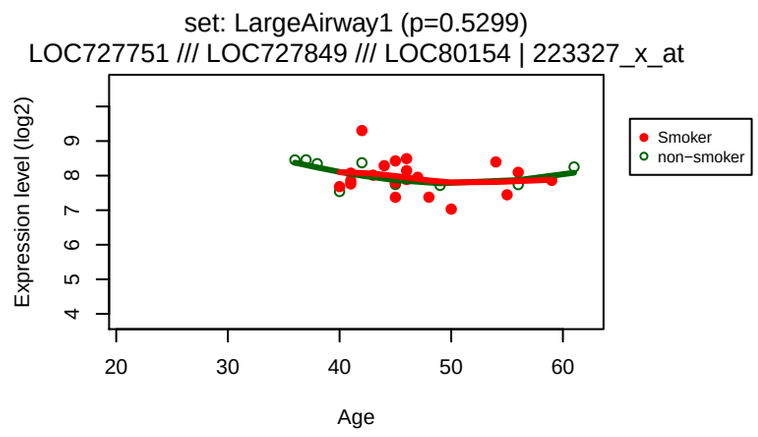

Gene expression of smokers and nonsmokers
gene: LOC727751 /// LOC727849 /// LOC80154 | 223327_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1093
p=0.9818
p=0.7473
p=0.5299
p=0.1095
p=0.3185
p=0.6942
p=0.6263
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

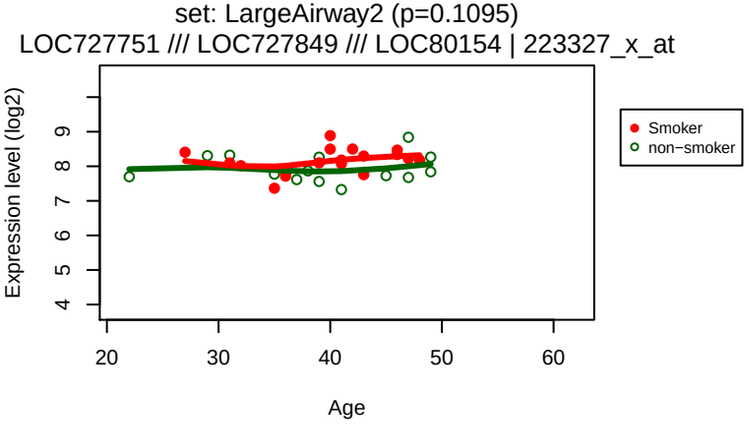

Gene expression ratio (smokers/nonsmokers)
gene: LOC727751 /// LOC727849 /// LOC80154 | 223327_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1093
p=0.9818
p=0.7473
p=0.5299
p=0.1095
p=0.3185
p=0.6942
p=0.6263
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

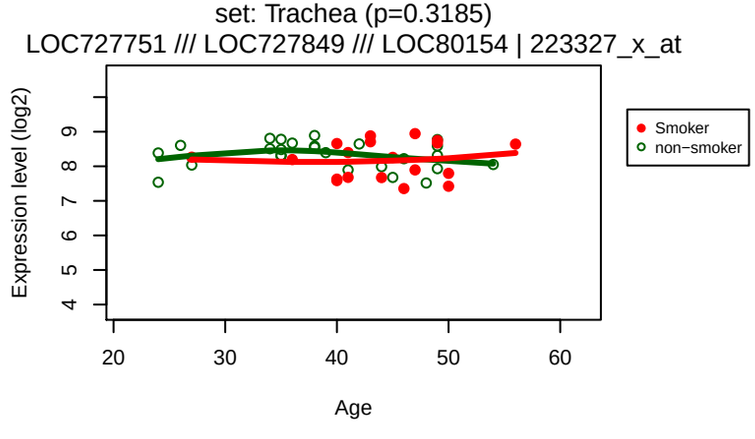

Gene expression ratio in Males
gene: LOC727751 /// LOC727849 /// LOC80154 | 223327_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7915
p=0.7647
p=0.1244
p=0.9103
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC727751 /// LOC727849 /// LOC80154 | 223327_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.993
p=0.004
p=0.8476
p=0.2172
SmallAirway2
LargeAirway2
Trachea
Macrophage2

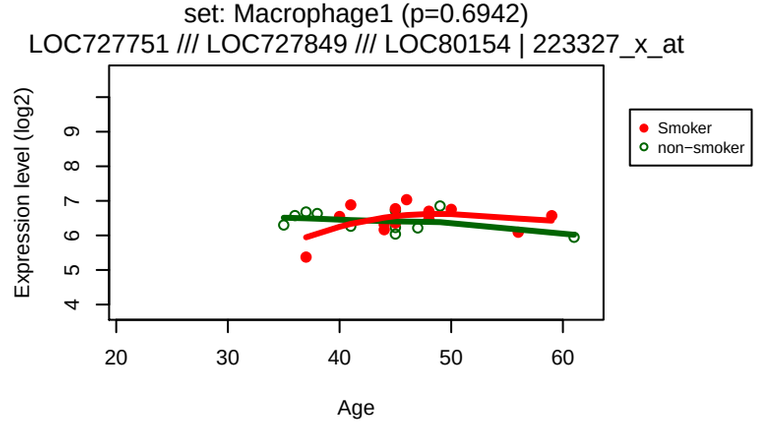

set: SmallAirway1 (p=0.1093)
LOC727751 /// LOC727849 /// LOC80154 | 223327_x_at
Expression level (log2)
Age
Smoker
non−smoker

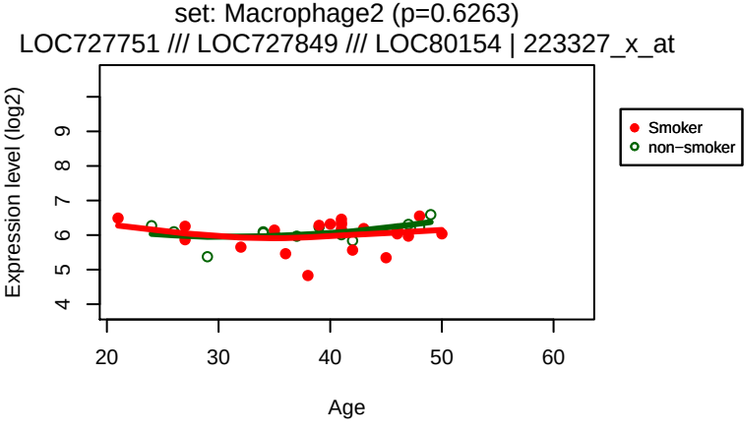

set: SmallAirway2 (p=0.9818)
LOC727751 /// LOC727849 /// LOC80154 | 223327_x_at
Expression level (log2)
Age
Smoker
non−smoker

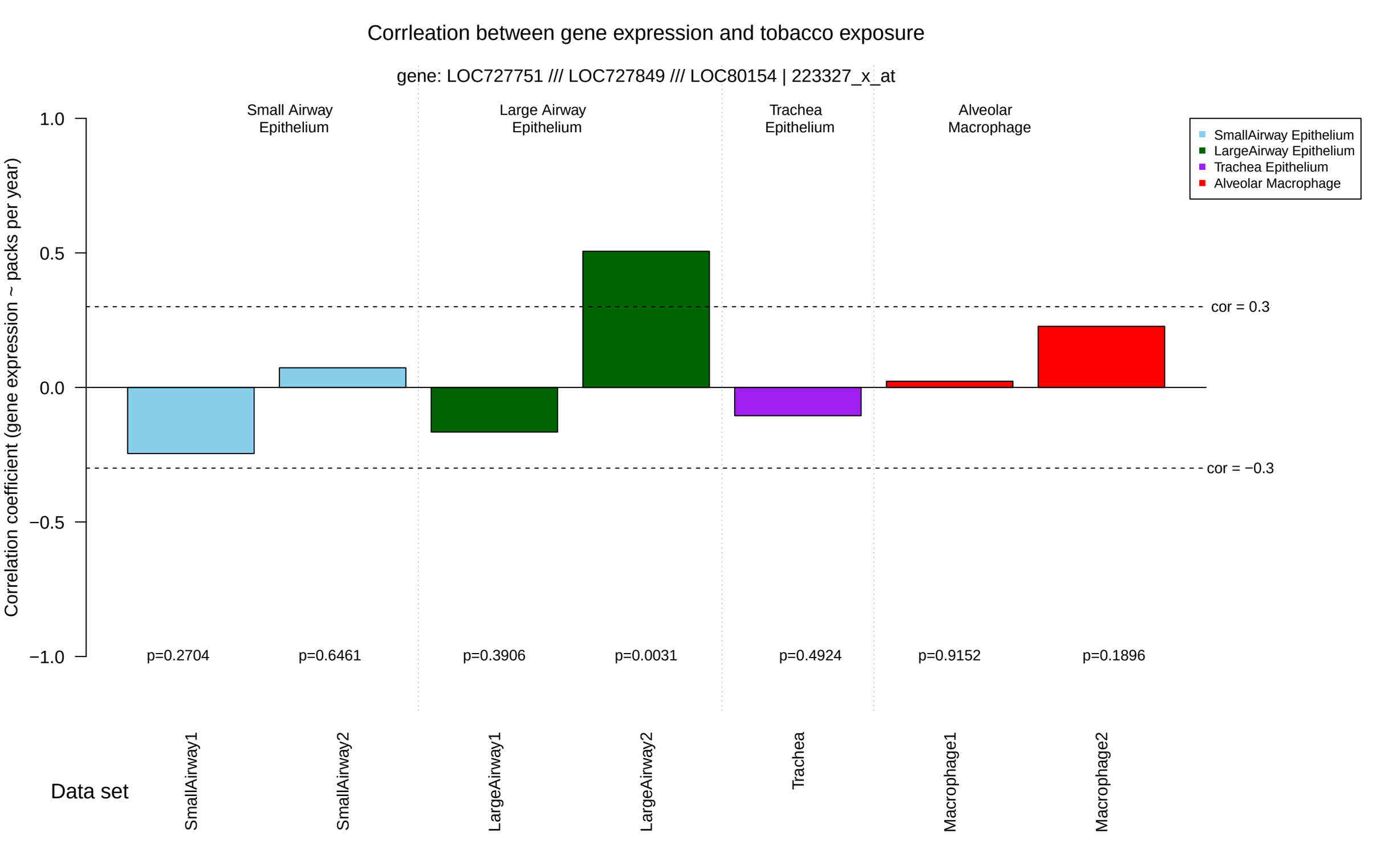

set: LargeAirway1 (p=0.5299)
LOC727751 /// LOC727849 /// LOC80154 | 223327_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1095)
LOC727751 /// LOC727849 /// LOC80154 | 223327_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3185)
LOC727751 /// LOC727849 /// LOC80154 | 223327_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6942)
LOC727751 /// LOC727849 /// LOC80154 | 223327_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6263)
LOC727751 /// LOC727849 /// LOC80154 | 223327_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC727751 /// LOC727849 /// LOC80154 | 223327_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2704
p=0.6461
p=0.3906
p=0.0031
p=0.4924
p=0.9152
p=0.1896
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2